UNIVERSITY OF CAPE COAST

DEPARTMENT OF MATHEMATICS AND STATISTICS

End of First Semester Examinations (2010/2011)

MAT 101: ALGEBRA AND TRIGONOMETRY

Time Allowed: 2 hours

Answer all **thirty (30)** questions in Section A, which carries 50% of the total marks, and **any two (2)** questions in Section B.

SECTION A

1. If logarithm of any number is to *base e* what is it called?

   (a) Standard logarithm (b) Common logarithm

   (c) Natural logarithm (d) Exponential logarithm

2. Which of the following is an odd function?

   (a) $x^3 - 5x + 3$

   (b) $x^3 + x^2 - 6x$

   (c) $4x^3 - 5x$

   (d) $6x^4 + 3x^2 + 1$

3. Which of the following forms would the partial fraction decomposition of $\frac{x^3+2x-1}{(x^2-4)^2(x^2+1)}$ represent?

   (a) $\frac{Ax+B}{x^2-4} + \frac{Cx+D}{(x^2-4)^2} + \frac{Ex+F}{x^2+1}$

   (b) $\frac{A}{x-2} + \frac{B}{(x-2)^2} + \frac{C}{x+2} + \frac{D}{(x+2)^2} + \frac{Ex+F}{x^2+1}$

   (c) $\frac{A}{(x-2)^2} + \frac{B}{(x+2)^2} + \frac{Cx+D}{x^2+1}$

   (d) $\frac{A}{x-2} + \frac{B}{x+2} + \frac{C}{x-1} + \frac{D}{x+1}$

4. Find the vertical asymptote (if any exists) for the curve of the function $f(x) = \frac{x^3-16}{x-4}$

   (a) $x = -4$

   (b) $x = 4$

   (c) $y = x + 4$

   (d) No vertical asymptote exists.

5. Find the solution set of $\begin{vmatrix} 3-x & x+1 \\ 5-x & x+2 \end{vmatrix} = -6$

   (a) $x = \frac{7}{3}$ (b) $x = 1$ (c) $x = \frac{5}{3}$ (d) $x = \frac{7}{5}$

6. A function is given by $f(x) = |x| - 4$. What is the range of $f(x)$

   (a) $\{y \mid y > -4\}$ (b) $\{y \mid y \geq 4\}$ (c) $\{y \mid y > 4\}$ (d) $\{y \mid y \geq -4\}$

7. Solve $\frac{x+6}{4} - \frac{x}{3} < 9$

   (a) $x > 62$

   (b) $x < 62$

   (c) $x < 106$

   (d) $x > 106$

8. Find the determinant of the matrix $A = \begin{bmatrix} 1 & 3 & -2 \\ 4 & 1 & -1 \\ 5 & -3 & 2 \end{bmatrix}$

   (a) -12 (b) -6 (c) 6 (d) 12

9. Given that $A = \begin{pmatrix} 1 & 3 & -2 \\ 4 & 1 & -1 \\ 5 & -3 & 2 \end{pmatrix}$, determine the (3,2)- cofactor of A.

   (a) -11 (b) -6 (c) 6 (d) 11

10. Which of the following is not a subset of Q?

    (a) Z (b) W (c) R (d) N

11. If $\log 2 = 0.3010$ *and* $\log 3 = 0.4770$, *find* $\log \frac{1}{7.5}$.

    (a) -1.875 (b) -0.875 (c) 0.875 (d) 1.875

12. Find the oblique asymptote for the curve $g(x) = \frac{x^2+5}{x+1}$.

    (a) $y = -6$ (b) $y = x - 1$ (c) $y = 6$ (d) $y = x + 1$

Use the information to answer questions 12 and 13

Given that $A = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}$ and $B = \begin{bmatrix} 5 & 4 \\ 0 & 2 \end{bmatrix}$

13. Determine the adjoint of $(A + B)$

(a) $\begin{pmatrix} 8 & -5 \\ -3 & -4 \end{pmatrix}$ (b) $\begin{pmatrix} -4 & 5 \\ 3 & 8 \end{pmatrix}$ (c) $\begin{pmatrix} 4 & 5 \\ 3 & 8 \end{pmatrix}$ (d) $\begin{pmatrix} 8 & 5 \\ 3 & 4 \end{pmatrix}$

14. Evaluate the inverse of matrix A

(a) 0 (b) $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

(c) $\begin{pmatrix} -4 & 5 \\ 3 & 8 \end{pmatrix}$ (d) the inverse does not exist

15. Solve $|2x + 1| > 7$

(a) $-3 < x < 4$ (b) $-4 < x < 3$

(c) $x < -3 \cup x > 4$ (d) $x < -4 \cup x > 3$

16. Decompose $\frac{1}{(x^2+5x+6)}$ into partial fractions

(a) $\frac{1}{x+2} + \frac{1}{x+3}$ (b) $\frac{1}{x+3} - \frac{1}{x+2}$

(c) $\frac{1}{x+2} - \frac{1}{x+3}$ (d) $-\frac{1}{x+2} - \frac{1}{x+3}$

17. Find the domain of the function, $g(x) = \sqrt{4 - x^2}$

(a) $-2 < x < 2$ (b) $0 < x < 2$

(c) $0 \leq x \leq 2$ (d) $-2 \leq x \leq 2$

18. Let A be an $n \times m$ matrix and B, an $n \times k$ matrix. What is the dimension of $A(A^T B)^n$

(a) $n \times m$ (b) $m \times m$ (c) $n \times k$ (d) $k \times m$

Use the information to answer questions 19 and 20

Given that the first term of an A P is 3 and the 20th term is 117

19. Find the common difference of the sequence

(a) 3 (b) 4 (c) 5 (d) 6

20. Determine the sum of the first 19 terms

(a) 111 (b) 112 (c) 113 (d) 114

21. Which of the following is a rational number, $\sqrt{0.16}, \pi, \sqrt{2}$ *and* $\sqrt{50}$.

(a) $\sqrt{0.16}$ (b) $\sqrt{2}$ (c) π (d) $\sqrt{50}$

22. Find the coefficient of the term $a^7 b^3$ in the expansion of $(a - 3b)^{10}$

(a) -3240 (b) -120 (c) 120 (d)3240

23. If $A = \{a, b\}, B = \{b, c\}$ *and* $C = \{d, e, f\}$, how many elements would be contained in the set $A \times B \times C$.

(a) 6 (b) 7 (c) 12 (d) 16

24. Find the values of x if ~~$4^{(x+\frac{1}{2})} + 4^{(\frac{3}{2}-x)} = 17$~~ $4^{(x+\frac{1}{2})} + 4^{(\frac{3}{2}-x)}$

(a) $x = -\frac{1}{2}, -\frac{3}{2}$ (b) $x = -\frac{1}{2}, \frac{3}{2}$ (c) $x = \frac{1}{2}, -\frac{3}{2}$ (d) $x = -\frac{1}{2}, -\frac{3}{2}$

25. Find the truth set of $\cos\theta = 0$ for $0 \leq \theta \leq 2\pi$.

(a) $\{\theta : \theta = 0, \frac{\pi}{2}\}$ (b) $\{\theta : \theta = 0, \frac{3\pi}{2}\}$

(c) $\{\theta : \theta = \frac{\pi}{2}, \frac{3\pi}{2}\}$ (d) $\{\theta : \theta = \frac{\pi}{2}, \frac{\pi}{4}\}$

26. Evaluate $\log(\log 1000^{1000})$

(a) 0 (b) 3.477 (c) [illegible] (d) [illegible]

27. Find the intercepts of the function $f(x) = x - 4$ on both axes

(a) $(0,-4),(0,4),(0,2)$ (b) $(-4,0),(4,0),(0,-4)$

(c) $(-4,0),(0,4),(0,4)$ (d) $(-4,0),(8,0),(0,-4)$

28. Solve the inequality $\left|\frac{x}{x+3}\right| \geq 3$

(a) $(-\infty,-3]$ or $[9,\infty)$ (b) $(-\infty,-3]$ or $(9,\infty)$

(c) $(-\infty,-3)$ or $(9,\infty)$ (d) $(-\infty,-3)$ or $[9,\infty)$

Use the information to answer questions 29 and 30

Let $U = \{Natural\ numbers\ less\ than\ 17\}$

$A = \{x \in U : x^2 - 11x + 28 = 0\}$

$B = \{x \in U : x\ is\ a\ factor\ of\ 12\}$

$C = \{3,6\}$

29. Find $(A \times B) \cap (C \times B)$

(a) $\emptyset$ (b) $\{(4,3),(4,6)\}$

(c) $\{(3,4),(6,4)\}$ (d) $\{(4,3),(7,3),(4,6),(7,6)\}$

30. How many elements would the set $(A \times B) \cap (A \times C)$ contain?

(a) 2 (b) 3 (c) 4 (d) 5

SECTION B

1. a) The notes of a piano, as measured in cycles per seconds, form a geometric sequence

i If A is 400 cycles per second and A' which has 12 nodes higher than A is 800cycles per second, find the constant ratio.

ii Find the cycles per second for C, three notes higher than A.

b) Solve for x if: $\log(x+3) + \log x = 1$

c) Simplify the expression $\frac{30\cos^3\theta\sin\theta}{6\sin^2\theta\cos\theta}$ assuming that $\sin\theta \neq 0$ and $\cos\theta \neq 0$.

2. a) Decompose $f(x) = \frac{x^3-7x^2+17x-17}{x^2-6x+9}$ into partial fractions.

b) If A, B and C are sets, show that:

$$A - (B \cap C) = (A - B) \cup (A - C)$$

c) Find all values of x in the interval $[0, 2\pi]$ such that $\sin x = \sin 2x$.